The elephant in the interpretability room: Why use attention as explanation when we have saliency methods?

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Abstract

There is a recent surge of interest in using attention as explanation of model predictions, with mixed evidence on whether attention can be used as such. While attention conveniently gives us one weight per input token and is easily extracted, it is often unclear toward what goal it is used as explanation. We find that often that goal, whether explicitly stated or not, is to find out what input tokens are the most relevant to a prediction, and that the implied user for the explanation is a model developer. For this goal and user, we argue that input saliency methods are better suited, and that there are no compelling reasons to use attention, despite the coincidence that it provides a weight for each input. With this position paper, we hope to shift some of the recent focus on attention to saliency methods, and for authors to clearly state the goal and user for their explanations.

1 Introduction

Attention mechanisms (Bahdanau et al., 2015) have allowed for performance gains in many areas of NLP, including, *inter alia*, machine translation (Bahdanau et al., 2015; Luong et al., 2015; Vaswani et al., 2017), natural language generation (e.g., Rush et al., 2015; Narayan et al., 2018), and natural language inference (e.g., Parikh et al., 2016).

Attention has not only allowed for better performance, it also provides a window into how a model is operating. For example, for machine translation, Bahdanau et al. (2015) visualize what source tokens the target tokens are attending to, often aligning words that are translations of each other.

Whether the window that attention gives into how a model operates amounts to *explanation* has recently become subject to debate (§2). While many papers published on the topic of explainable AI have been criticised for not defining explanations (Lipton, 2018; Miller, 2019), the first key studies which spawned interest in attention as explanation (Jain and Wallace, 2019; Serrano and Smith, 2019; Wiegreffe and Pinter, 2019) do say that they are interested in whether attention weights faithfully represent the responsibility each input token has on a model prediction. That is, the narrow definition of explanation implied there is that it points at the most important input tokens for a prediction (arg max), accurately summarizing the reasoning process of the model (Jacovi and Goldberg, 2020b).

The above works have inspired some to find ways to make attention more faithful and/or plausible, by changing the nature of the hidden representations attention is computed over using special training objectives (e.g., Mohankumar et al., 2020; Tutek and Snajder, 2020). Others have proposed replacing the attention mechanism with a latent alignment model (Deng et al., 2018).

Interestingly, the implied definition of explanation in the cited works, happens to coincide with what *input saliency methods* (§3) are designed to produce (Li et al., 2016a; Sundararajan et al., 2017; Ribeiro et al., 2016; Montavon et al., 2019, i.a.). Moreover, the user of that explanation is often implied to be a model developer, to which faithfulness is important. The elephant in the room is therefore: If the goal of using attention as explanation is to assign importance weights to the input tokens in a faithful manner, why should the attention mechanism be preferred over the multitude of existing input saliency methods designed to do *exactly that*? In this position paper, with that goal in mind, we argue that we should pay attention no heed (§4). We propose that we reduce our focus on attention as explanation, and shift it to input saliency methods instead. However, we do emphasize that understanding the *role* of attention is still a valid research goal (§5), and finally, we discuss a few approaches that go beyond saliency (§6).

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